

1. Connect the Monarch via USB cable to the host computer. The Monarch uses a USB-C connector, but the other end can be USB-A (most common) or any form that is convenient. The host computer needs only a USB 2.0 host connector. It can be a PC, an Apple, a Linux, or other platform. It can be a desktop, laptop, tablet, mobile device, Raspberry Pi, etc. You may use a hub. USB mode is CDC. The USB device should enumerate (connect) automatically. The Monarch uses a trusted USB driver, so there should never be a problem with the operating system. Make sure your USB cable is a complete data + power one, not a power-only one.

2. The Monarch will connect to a free COM port. Verify that its LED is flashing green slowly.

3. The default serial setting is 115,200 baud, 8N1, no hand-shaking. The default is to output strings in the example format below. Default output rate is 1 Hz. The magnetic measurement unit is μT . 1 Gauss = 100 μT . These data are raw, not filtered or averaged in any way. Note, the Earth's magnetic field varies widely and is not isotropic. The predominant vector will point toward magnetic north, but there will be a significant radial (z-axis) component. Local geology and structures will have a large effect. The Monarch's

measurements are immersed in this widely-varying and dynamic field.

4. You can communicate with the Monarch in the following ways:

- a. Through a terminal emulator program such as PuTTY or TeraTerm (both free). This alone may be sufficient for your needs.

- b. Through a desktop app, using Acorn commands. Acorn is a simple, efficient API for communicating with instruments. It is documented on the Black Oak Engineering [website](#). Monarch MMG3 Acorn support is listed herein below.

- c. The Monarch is often used in embedded systems. It simply needs a USB CDC port. Again, use the Acorn API to talk with it.

5. Avoid strong magnetic fields. 3 mT should be the limit. A rare earth magnet placed a few cm from the Monarch can easily exceed this limit.

Standard baud rate is 115200, 8N1.

All commands may be called with a '?' appended. This will give brief help on usage and range. E.g., 'ach?'.

Acquisition Commands

`ach` Acquisition Channel. What is the measurement in of the xyz axes?

`alt` Acquisition Latency Time. What is the basic measurement duration?

`amo` Acquisition Mode. Change operating mode. For factory only.

`aqp` Acquisition Period. What is the present acquisition period in ms?

`aqp=100` Set acquisition period in ms.

Note, in Monarch instruments, acquisition period is strictly equal to serial reporting period. There is no data storage, buffering, etc.

`asn` Acquisition Sample Number. Gives fixed sample number, if defined.

asn=20 Set fixed sample number to 20. Thus the Monarch will acquire 20 samples, then stop. Set to zero to restore continuous acquisition and reporting. asn# gives range. asn+ resets the counter.

Calibration Commands

bin Show calibration information.

bst Execute self test. For factory only.

This should be followed by a hard reset.

Device Commands

dco Point of contact information.

DCO='www.blackoakeng.com'

der Report any error conditions.

dmo Report model number.

dre Do a hard reset.

dsn Serial number?

dte What is the present coarse temperature?

dti What is the present system time?

dti='HH:MM' Set system time. Use 24 hr form of hours. Seconds and ms are reset. Note, Monarch uses a few different ms counters, some of which are 'free'. The latter do not count to 999; they count to 65,535. These may appear as the time stamps for sample reporting.

dtz Zero all ms timers.

dva What is your Acorn version?

dvf What is your firmware version?

dvh What is your hardware version?

dvn What is your device plain name?

Communications Commands

mal List all supported Acorn commands.

mco+|- Enable | disable serial feed.

mec+|- Enable | disable command echo.

mhd Show column headers for data lines.